

Housing Price Analysis Report

Introduction

This report provides a data-driven analysis of the residential housing market using the classic [Kaggle Housing Prices Dataset](#). The dataset consists of 545 individual housing records evaluated across 13 distinct features. It combines numeric structural dimensions such as total square footage (area), bedroom counts, bathrooms, and stories with categorical indicators of premium amenities like air conditioning, basement, and location preferences. The main goal of this project is to find out exactly how a house's size, features, and location change its price, so that buyers can get the best possible value for their money.

Key Findings and Insights

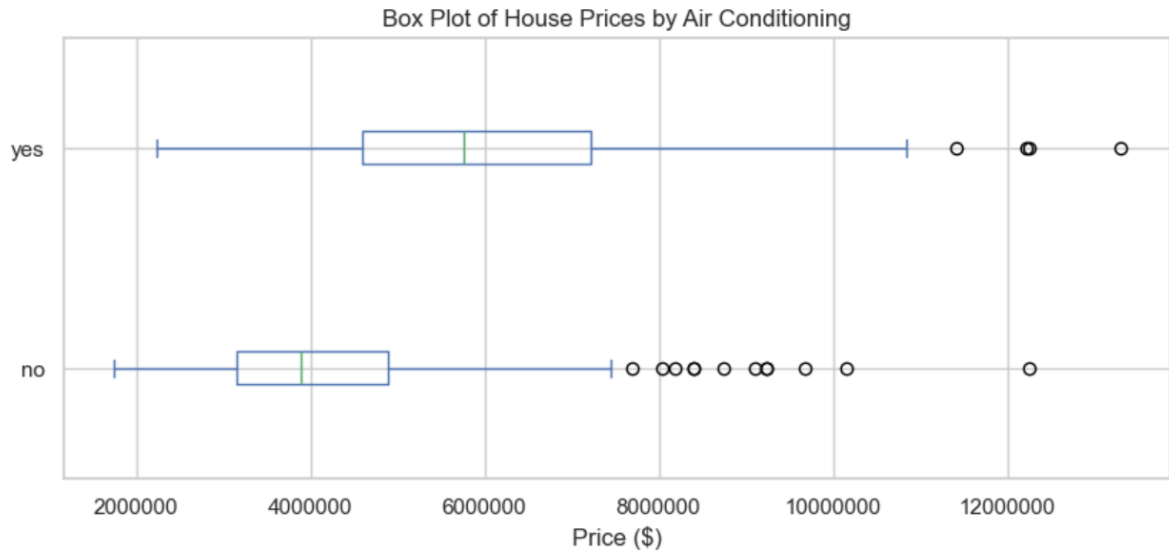
By analyzing the data through averages, groupings, and relationship scores, we uncovered four major insights about house traits and prices:

- **The High Price of Air Conditioning:** Having air conditioning vastly increases a home's value. The data shows a huge price gap, where homes with air conditioning cost much more on average than homes that rely only on natural ventilation.
- **The Premium Value of Furnishing:** The way a house is furnished directly changes its rank in the market. Fully furnished homes command the highest average price, followed by semi-furnished homes, while completely unfurnished properties sit at the very bottom of the price ladder.
- **The Power of Combining Features:** Adding great features together drastically spikes a home's value. Specifically, our data shows that exactly 62 homes have both a basement and air conditioning, and this premium group commands a high average price of 6,081,196.97.
- **The 3-Bedroom Sweet Spot:** When looking at how many bedrooms these houses have, one specific style completely dominates the market. Exactly 55.05% of all the homes have exactly 3 bedrooms, making this layout the most common and easiest type to find.

Visualizations

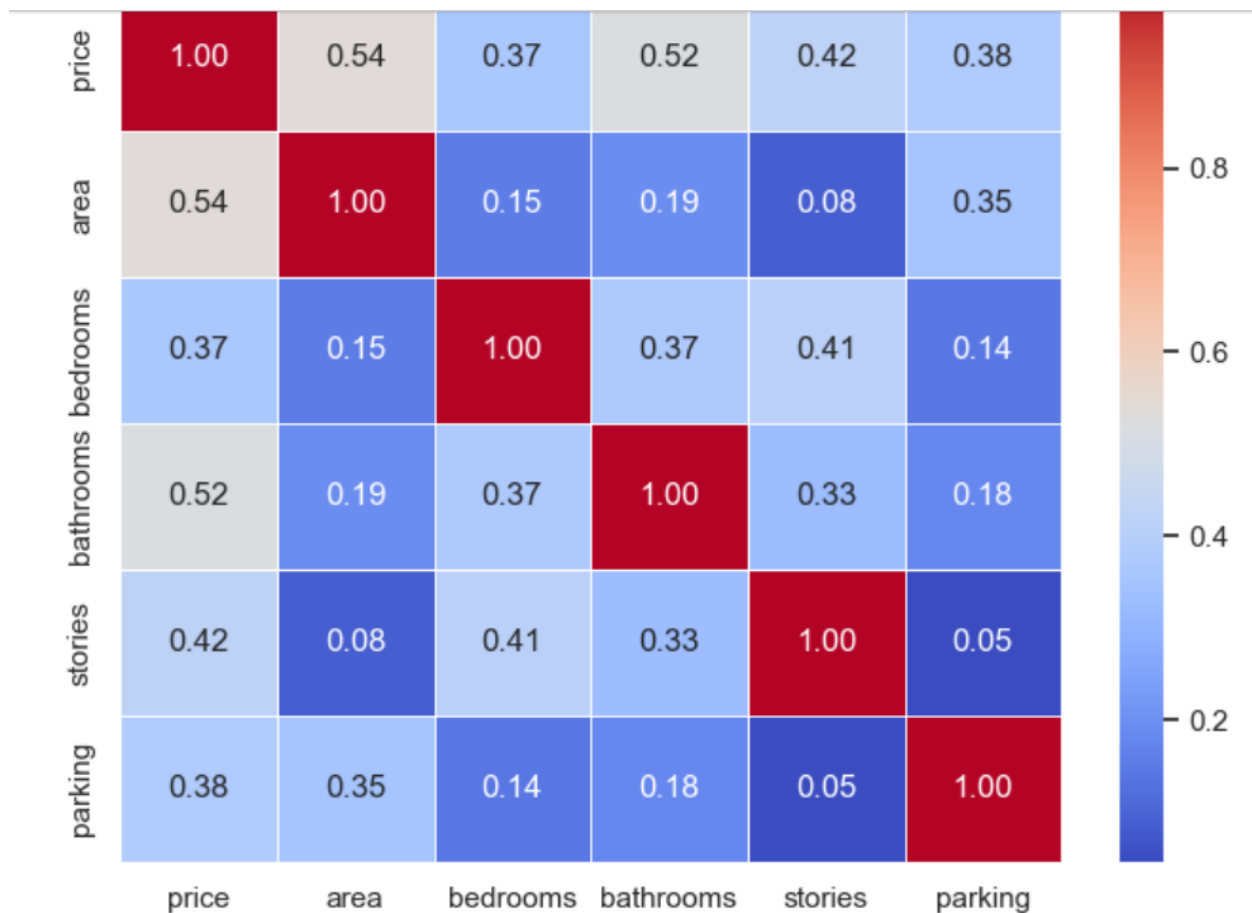
To validate these statistical findings visually, two primary charts were rendered:

Visual Component 1: Box Plot of Price Grouped by Air Conditioning



- **Interpretation:** This chart contrasts the price spreads of homes with and without AC. The box corresponding to 'yes' is significantly shifted to the right with a much higher median line, visually confirming that air conditioning consistently elevates the market value of properties.

Visual Component 2: Correlation Heatmap of Numeric Variables



- Interpretation:** This correlation grid maps out the linear strength between variables. It visually flags the moderate 0.42 correlation between stories and price, alongside revealing that total square footage (area) maintains the strongest overall linear impact on a home's final valuation.

Conclusion and Strategic Recommendations

In conclusion, residential market values are intensely dictated by a mix of transit accessibility, layout, and high-demand interior utilities. For a hypothetical home buyer looking to maximize absolute value per currency spent, the following three actionable data-backed recommendations are proposed:

- Target Off-Main-Road Properties:** Look for homes located off primary roads. This single choice saves buyers an average of over 1.5 million in premium costs while frequently maintaining identical interior square footage.

2. **Prioritize 3-Bedroom Inventory:** Leverage the 55.05% market saturation of 3-bedroom layouts to negotiate aggressive discounts. The sheer volume of this inventory reduces seller leverage compared to rarer layout options.
3. **Sacrifice Excess Vertical Stories:** Opt for a spacious 1 or 2-story home rather than a 3- or 4-story layout. Given the moderate 0.42 correlation, verticality adds an exponential luxury premium that often overshadows the practical value of the actual underlying ground area.